DG INTERNS HUB

AI/ML INTRESHIP -WEEK 2 TASKS

- **TOPIC**: Data Handling using NumPy and pandas.
- **Tools Used**: python, NumPy, Pandas, Visual Studio code.
- **Dataset**: Student Performance Dataset (CSV).

- **Submitted By**: Neha Fule.
- **Internship Domain**: Artificial intelligence/ Machine learning.
- **Submission Date**: 14 October 2026

Task 1: Dataset Loading & Understanding.

- To load and understand the csv dataset using pandas,
- The dataset contains 10 student records, Roll no , Name, Python , Java, DBMS, and Department
- **Code:**

```
import pandas as pd

df = pd.read_csv("Week2/student.csv")

print("FIRST 5 ROWS")

print(df.head())

print("\nLAST 5 ROWS")

print(df.tail())

print("\nDATASET SHAPE")

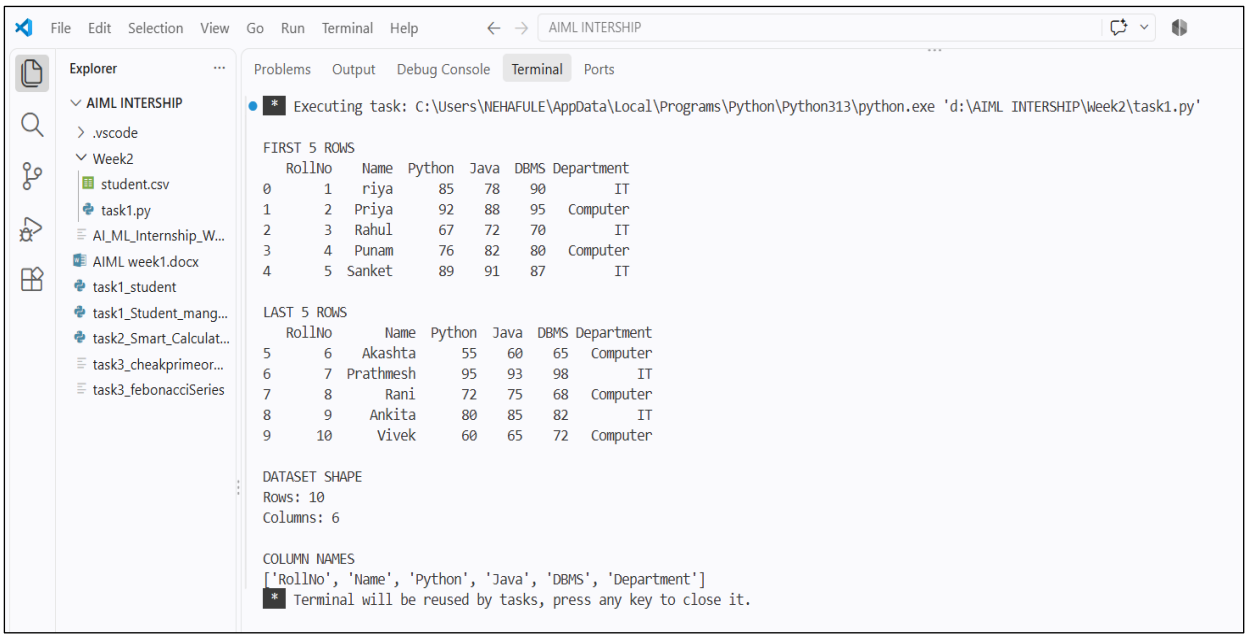
print("Rows:", df.shape[0])

print("Columns:", df.shape[1])

print("\nCOLUMN NAMES")

print(df.columns.tolist())
```

- **output:**



Task 2 – Data Cleaning

- To identify and handle missing values and duplicate records.
- Checked missing values using isnull().sum().
- Checked duplicate records using duplicated().sum().
- Filled missing numerical values using column mean.
- Removed duplicate rows using drop_duplicates().
- Saved the cleaned dataset as cleaned_students.csv.

Code:

```
import pandas as pd

df = pd.read_csv("Week2/student.csv")

print("MISSING VALUES")

print(df.isnull().sum())

print("\nDUPLICATE ROWS")

print(df.duplicated().sum())

numeric_cols = ["Python", "Java", "DBMS"]

df[numeric_cols] = df[numeric_cols].fillna(

   df[numeric_cols].mean()

)

df = df.drop_duplicates()

print("\nDATA AFTER CLEANING")

print(df)

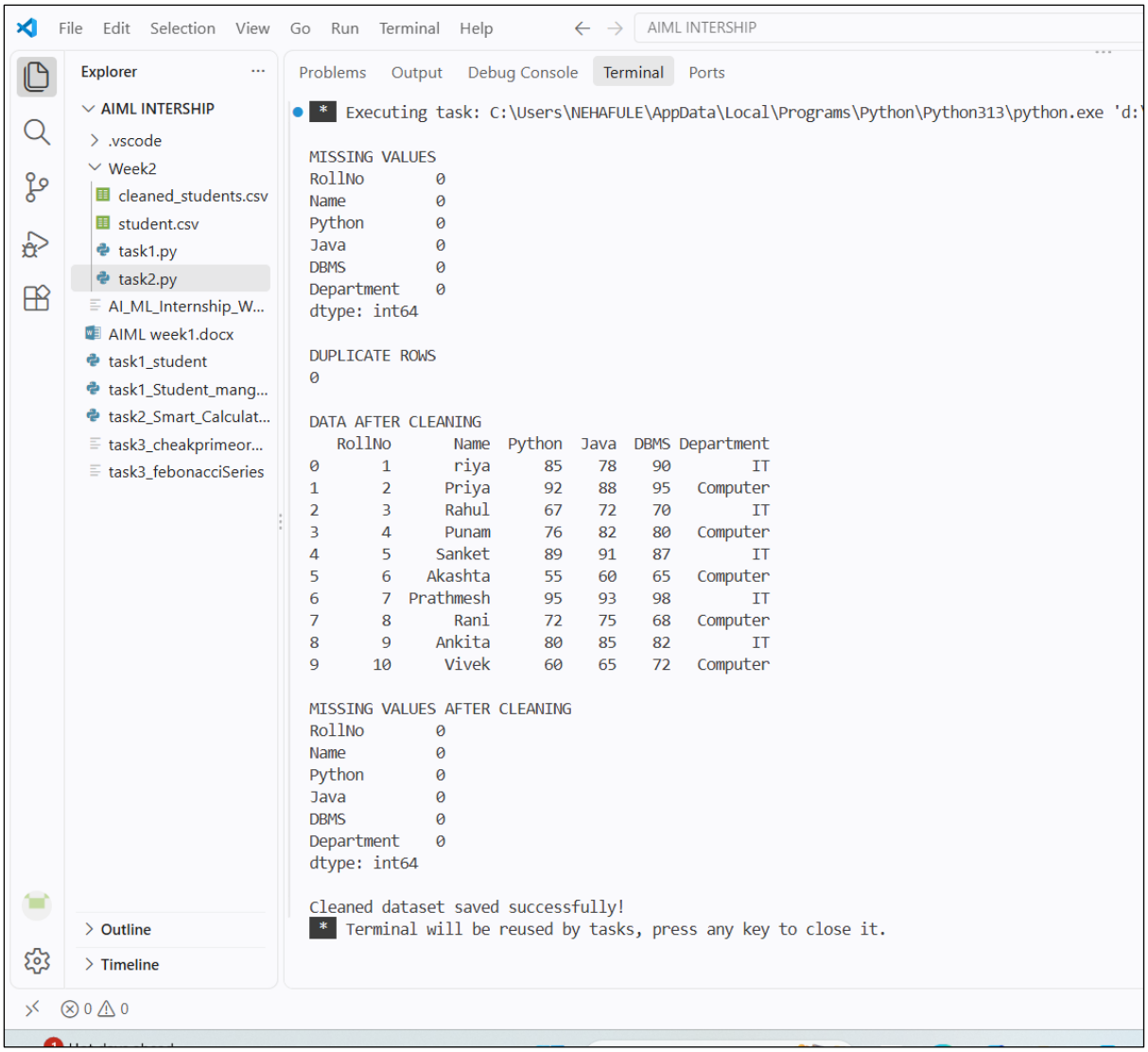
print("\nMISSING VALUES AFTER CLEANING")

print(df.isnull().sum())

df.to_csv("Week2/cleaned_students.csv", index=False)

print("\nCleaned dataset saved successfully!")
```

output:



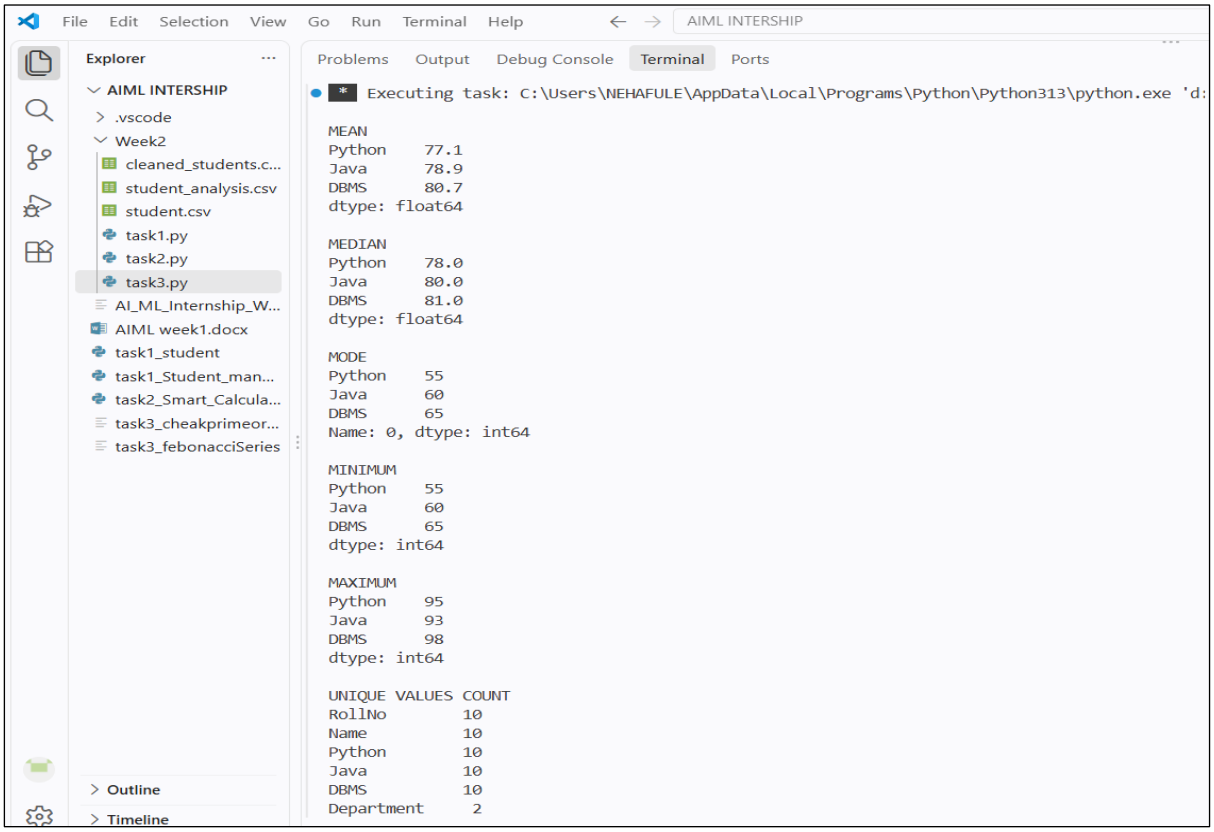
Task 3 – Data Analysis

- To calculate statistical measures and understand the dataset.
    - Mean
    - Median
    - Mode
    - Minimum and Maximum
    - Count of unique values
    - Most frequent department
    - Average marks of each student

Code:

```
import pandas as pd
df = pd.read_csv("Week2/cleaned_students.csv")
subjects = ["Python", "Java", "DBMS"]
print("MEAN")
print(df[subjects].mean())
#mode
print("\nMEDIAN")
print(df[subjects].median())
#mean
print("\nMODE")
print(df[subjects].mode().iloc[0])
# Minimum
print("\nMINIMUM")
print(df[subjects].min())
# Maximum
print("\nMAXIMUM")
print(df[subjects].max())
# Count of unique values
print("\nUNIQUE VALUES COUNT")
print(df.nunique())
# Most frequent department
print("\nMOST FREQUENT DEPARTMENT")
print(df["Department"].mode()[0])
df["Average"] = df[subjects].mean(axis=1)
print("\nSTUDENT AVERAGE MARKS")
print(df[["Name", "Average"]])
df.to_csv("Week2/student_analysis.csv", index=False)
print("\nAnalysis completed successfully!")
```

output:




```
UNIQUE VALUES COUNT
RollNo         10
Name           10
Python         10
Java           10
DBMS           10
Department      2
dtype: int64

MOST FREQUENT DEPARTMENT
Computer

STUDENT AVERAGE MARKS
        Name    Average
0       riya  84.333333
1      Priya  91.666667
2      Rahul  69.666667
3      Punam  79.333333
4     Sanket  89.000000
5    Akashta  60.000000
6  Prathmesh  95.333333
7       Rani  71.666667
8     Ankita  82.333333
9      Vivek  65.666667

Analysis completed successfully!
* Terminal will be reused by tasks, press any key to close it.
```

Task 4 – NumPy Practice

- To perform basic numerical operations using NumPy arrays.

- Created two NumPy arrays.
- Performed array addition.
- Performed array multiplication.
- Reshaped an array into a 2 × 3 matrix.

```
import numpy as np

arr1 = np.array([10, 20, 30, 40, 50, 60])

arr2 = np.array([1, 2, 3, 4, 5, 6])

print("ARRAY 1:")

print(arr1)

print("\nARRAY 2:")

print(arr2)                    # Addition

print("\nADDITION:")

print(arr1 + arr2)    # Multiplication

print("\nMULTIPLICATION:")

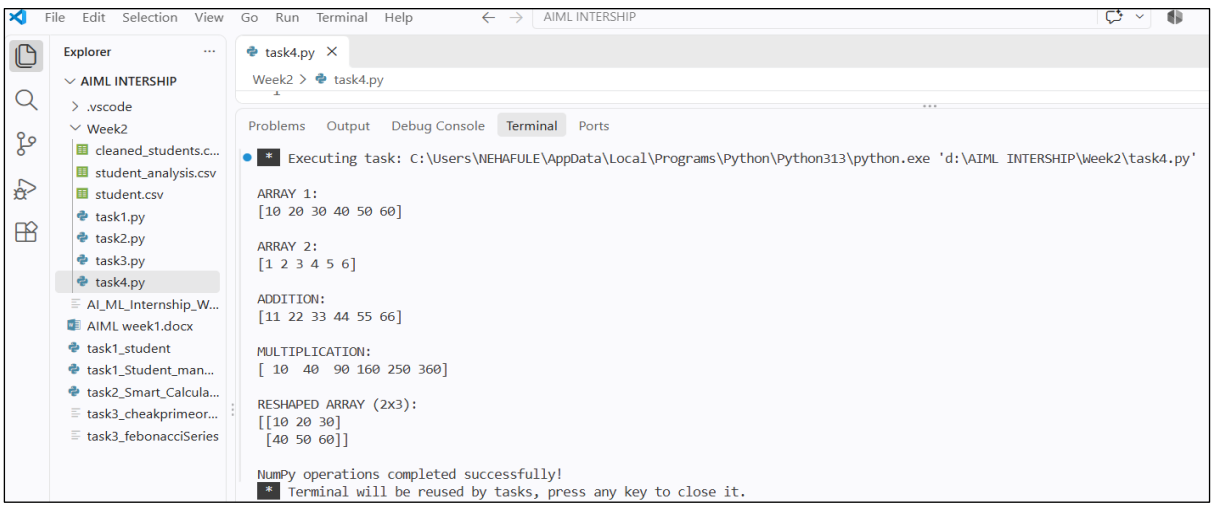
print(arr1 * arr2)                  # Reshape array

print("\nRESHAPED ARRAY (2x3):")

print(arr1.reshape(2, 3))

print("\nNumPy operations completed successfully!")
```

output:



➢ **Bonus Task – Data Filtering and Sorting**

- To filter records and display the top 10 records.
- Filtered students whose Python marks were greater than 80.
- Displayed the top 10 records.
- Sorted students by Python marks in descending order.

```
import pandas as pd

df = pd.read_csv("Week2/cleaned_students.csv")

print("STUDENTS WITH PYTHON MARKS > 80")

filtered_df = df[df["Python"] > 80]

print(filtered_df)

print("\nTOP 10 RECORDS")

print(df.head(10))

print("\nSTUDENTS SORTED BY PYTHON MARKS")

print(df.sort_values(by="Python", ascending=False))

print("\nBonus task completed successfully!")
```

output:

